

DATA FUNDAMENTALS PROJECT

TITLE - Retail Store Sales Analysis

By -:

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Retail Store Sales Analysis

Problem Statement

Retail company unable to identify revenue drivers and sales trends.

Solution

- Data analysis using SQL, Excel, Python
- Dashboard visualization
- Insights for decision making

Features

- Revenue analysis
- Product performance
- City comparison
- Trend visualization

OUTPUT

SQL output

Display all records

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	order_id	order_date	city	product	category	quantity	price
▶	1001	2025-01-10	Pune	Laptop	Electronics	2	50000
	1002	2025-01-11	Mumbai	Mobile	Electronics	3	20000
	1003	2025-01-11	Pune	Headphones	Accessories	5	2000
	1004	2025-01-12	Nagpur	Keyboard	Accessories	4	1500
	1005	2025-01-13	Mumbai	Laptop	Electronics	1	50000
	1006	2025-01-14	Pune	Mobile	Electronics	2	22000
	1007	2025-01-15	Nagpur	Laptop	Electronics	1	52000
	1008	2025-01-30	Mumbai	Headphones	Accessories	6	1800
	1009	2025-01-02	Bangalore	Mobile	Electronics	3	22000
	1010	2025-01-28	Guwahati	Laptop	Electronics	4	52000
	1011	2025-01-22	Ahmedabad	Headphones	Accessories	10	1800

Total Revenue

```
32
33 # Total revenue
34
35 • SELECT SUM(quantity * price) AS total_revenue FROM sales;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
total_revenue			
▶ 624800			

Top-selling product (by quantity)

```
36
37 # Top-selling product (by quantity)
38
39 • SELECT product, SUM(quantity) AS total_qty FROM sales GROUP BY product ORDER BY total_qty DESC LIMIT 1;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
product				
▶ Headphones				21

Revenue by city

```
41 # Revenue by city
42
43 • SELECT city, SUM(quantity * price) AS revenue FROM sales GROUP BY city;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
city			
▶ Pune			154000
Mumbai			120800
Nagpur			58000
Bangalore			66000
Guwahati			208000
Ahmedabad			18000

Category-wise revenue

44

45 # Category-wise revenue

46

47 • `SELECT category, SUM(quantity * price) AS revenue FROM sales GROUP BY category;`

48

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	category	revenue			
▶	Electronics	580000			
	Accessories	44800			

Highest value order

49 # Highest value order

50

51 • `SELECT *, (quantity * price) AS total FROM sales ORDER BY total DESC LIMIT 1;`

52

Result Grid									Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
	order_id	order_date	city	product	category	quantity	price	total				
▶	1010	2025-01-28	Guwahati	Laptop	Electronics	4	52000	208000				

Average product price

53 # Average product price

54

55 • `SELECT AVG(price) AS avg_price FROM sales;`

56

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	avg_price
▶	25009.0909

Excel pivot

Total sales

order_id	order_date	city	product	category	quantity	price	Total Sales
1001	10-01-2025	Pune	Laptop	Electronics	2	50000	100000
1002	11-01-2025	Mumbai	Mobile	Electronics	3	20000	60000
1003	11-01-2025	Pune	Headphones	Accessories	5	2000	10000
1004	12-01-2025	Nagpur	Keyboard	Accessories	4	1500	6000
1005	13-01-2025	Mumbai	Laptop	Electronics	1	50000	50000
1006	14-01-2025	Pune	Mobile	Electronics	2	22000	44000
1007	15-01-2025	Nagpur	Laptop	Electronics	1	52000	52000
1008	16-01-2025	Mumbai	Headphones	Accessories	6	1800	10800
1009	02-01-2025	Bangalore	Mobile	Electronics	3	22000	66000
1010	28-01-2025	Guwahati	Laptop	Electronics	4	52000	208000
1011	22-01-2025	Ahmedabad	Headphones	Accessories	10	1800	18000

Total sales (desc)

	A	B	C	D	E	F	G	H
1	order_id	order_date	city	product	category	quantity	price	Total Sales
2	1010	28-01-2025	Guwahati	Laptop	Electronics	4	52000	208000
3	1001	10-01-2025	Pune	Laptop	Electronics	2	50000	100000
4	1009	02-01-2025	Bangalore	Mobile	Electronics	3	22000	66000
5	1002	11-01-2025	Mumbai	Mobile	Electronics	3	20000	60000
6	1007	15-01-2025	Nagpur	Laptop	Electronics	1	52000	52000
7	1005	13-01-2025	Mumbai	Laptop	Electronics	1	50000	50000
8	1006	14-01-2025	Pune	Mobile	Electronics	2	22000	44000
9	1011	22-01-2025	Ahmedabad	Headphones	Accessories	10	1800	18000
10	1008	16-01-2025	Mumbai	Headphones	Accessories	6	1800	10800
11	1003	11-01-2025	Pune	Headphones	Accessories	5	2000	10000
12	1004	12-01-2025	Nagpur	Keyboard	Accessories	4	1500	6000

Filter (City)

	A	B	C	D	E	F	G	H
1	order_id	order_date	city	product	category	quantity	price	Total Sales
3	1001	10-01-2025	Pune	Laptop	Electronics	2	50000	100000
8	1006	14-01-2025	Pune	Mobile	Electronics	2	22000	44000
11	1003	11-01-2025	Pune	Headphones	Accessories	5	2000	10000
13								
14								
15								

Pivot Table

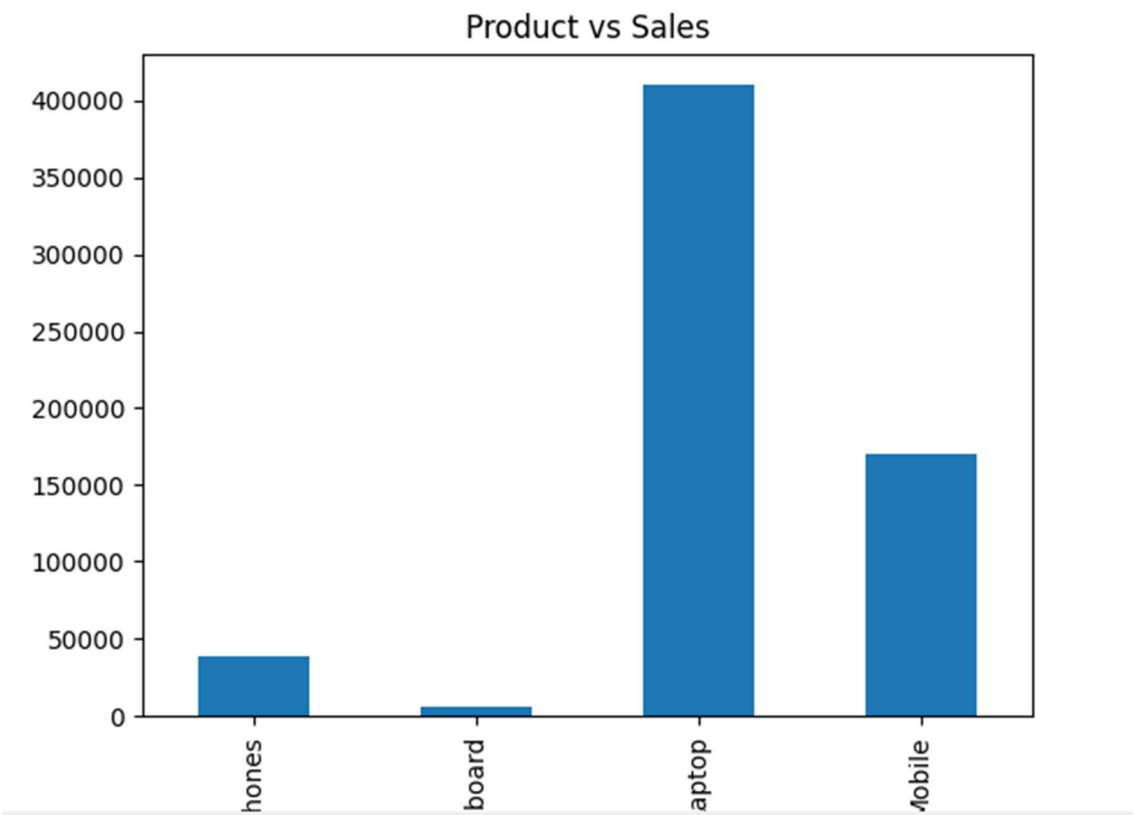
16		
17	Row Labels	Sum of Total Sales
18	• Headphones	38800
19	Ahmedabad	18000
20	Mumbai	10800
21	Pune	10000
22	• Keyboard	6000
23	Nagpur	6000
24	• Laptop	410000
25	Guwahati	208000
26	Mumbai	50000
27	Nagpur	52000
28	Pune	100000
29	• Mobile	170000
30	Bangalore	66000
31	Mumbai	60000
32	Pune	44000
33	Grand Total	624800
34		
35		

Best product => Laptop

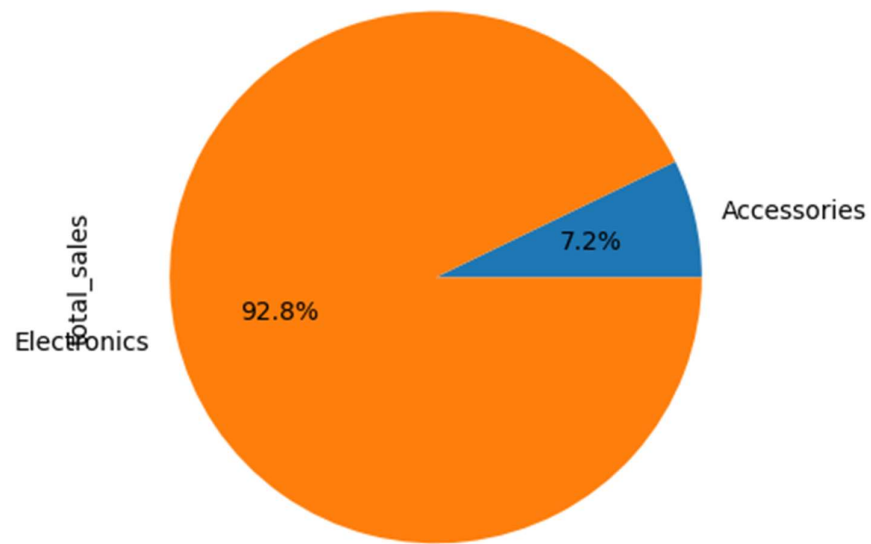
Best city => Mumbai

Electronics => Accessories

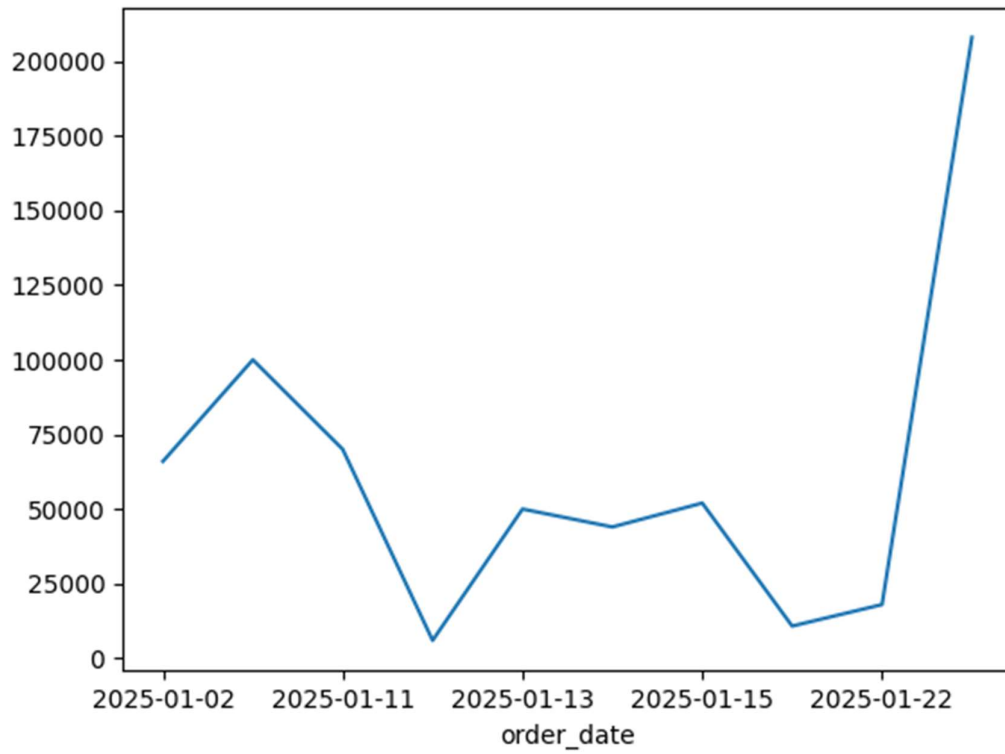
Python charts



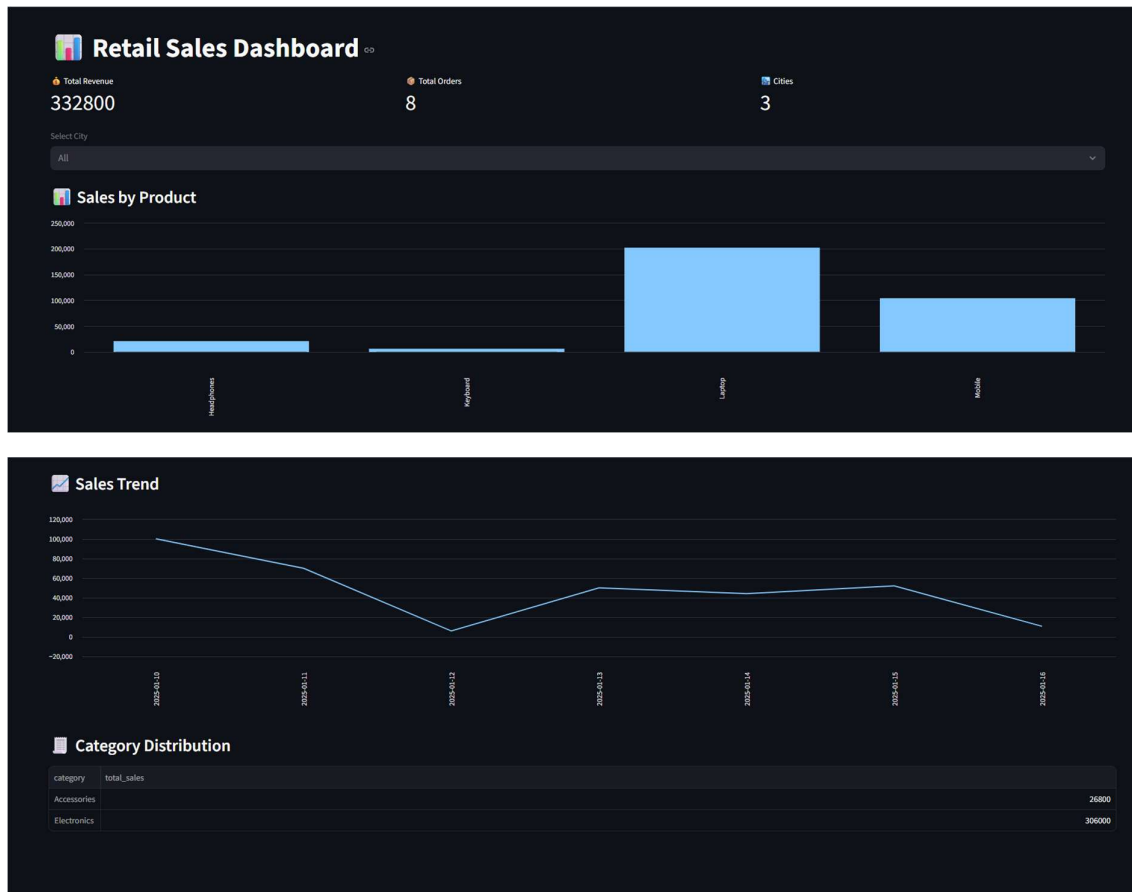
Category Distribution



Sales Over Time



Dashboard



Tech Stack

- Python (Pandas, Matplotlib)
- SQL (MySQL)
- Excel
- Streamlit

Unique Points

- Multi-tool analysis (SQL + Excel + Python)
- Interactive dashboard
- Real-world business insights

Future Improvements

- Add real-time data API
- Add ML prediction
- Convert to web app